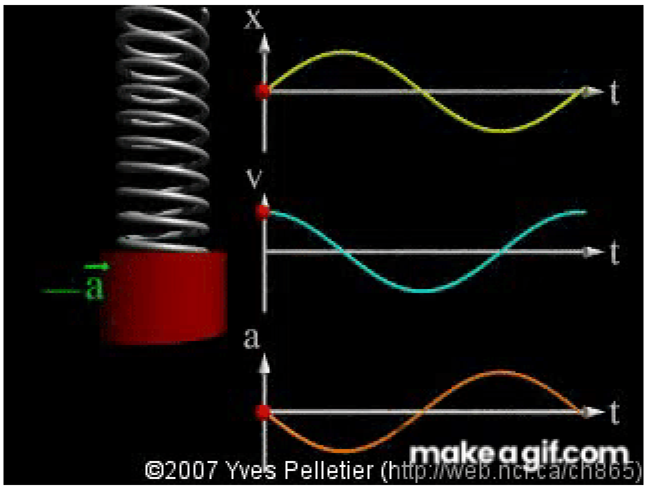
popquiz simple harmonic motion, elastic PE, Hooke's law - BONUS

This is a preview of the published version of the quiz

Started: Apr 29 at 7:39pm

Quiz Instructions



Question 1 14.5 pts

go to: **https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html** **(https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html)**

click on LAB

attach a 100g mass to the spring. Select no damping. Select slow. Select velocity (V) and acceleration (a). Pull on the mass and let it oscillate. Observe the vectors a and V (green and yellow)

- ○ Normal
- ● Slow

When the mass is at its lowest position what is true (or highest)

note: the stretch is maximum.

○ a=0 and V =0

- Can not be determined
- a is max and V is max
- a is max and V=0
- V is max and a =0

Question 2 14.5 pts

Building on the previous question. observe vectors a and V as the mass oscillates. What is true?

Let's called F the restoring force (hookes law). " You shall not forget Newton's second law : a = .. /m"

- a=0 and F is a variable force.
- a is not contant but F is constant
- a is not constant, F is not constant
- a is constant but F is a variable force

Question 3 14.5 pts

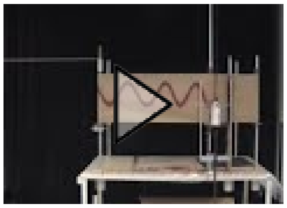
Building on the previous questions. When the mass is going through its rest position (when the spring is not stretched/compressed) while still moving, what is true:

- V is max and a =0
- V=0 and a =0
- Both a and V are max
- V=0 and a is max

Question 4 14.5 pts

watch this demo with a harmonic oscillator:

https://www.youtube.com/watch?v=P-Umre5Np_0 (https://www.youtube.com/watch?v=P-Umre5Np_0)



[(https://www.youtube.com/watch?v=P-Umre5Np_0)](https://www.youtube.com/watch?v=P-Umre5Np_0)

The vertical displacement of the oscillator as a function of time describes

- a straight line
- a parabola
- a sine wave
- an exponential
- a horizontal line

Question 5 20 pts

https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html
(https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html)

Go to energy. (click on the tab))

select no damping and hang a mass. Pull on the mass so it oscillates. click on slow and focus of KE and PE elastic (stored in spring) so PEelastic and KE.

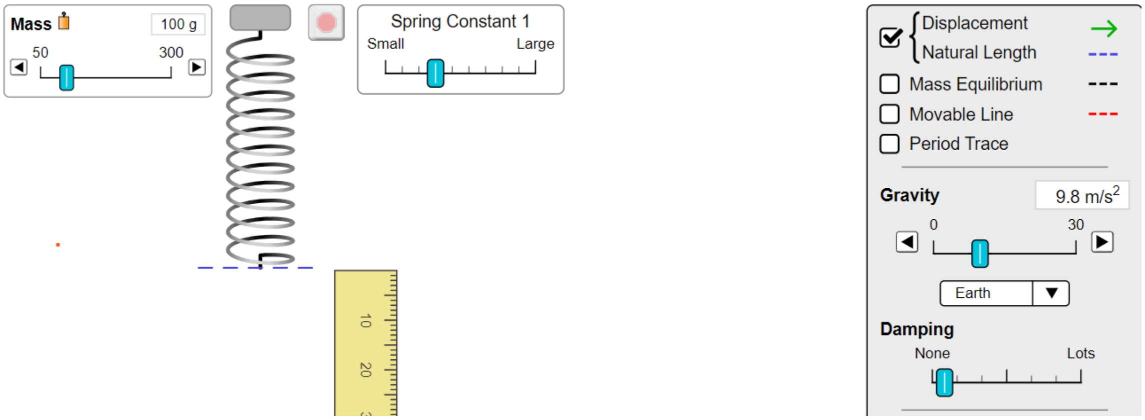
- Energy is not swinging back between KE and PE
- When KE is max, PE is max
- when PE is max, KE is 0
- When PE is 0, KE is 0

Question 6 14.5 pts

Lets go back to:

[https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html (https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html)](https://phet.colorado.edu/sims/html/masses-and-springs/latest/masses-and-springs_all.html)

Select Lab. Click on displacement green arrow.



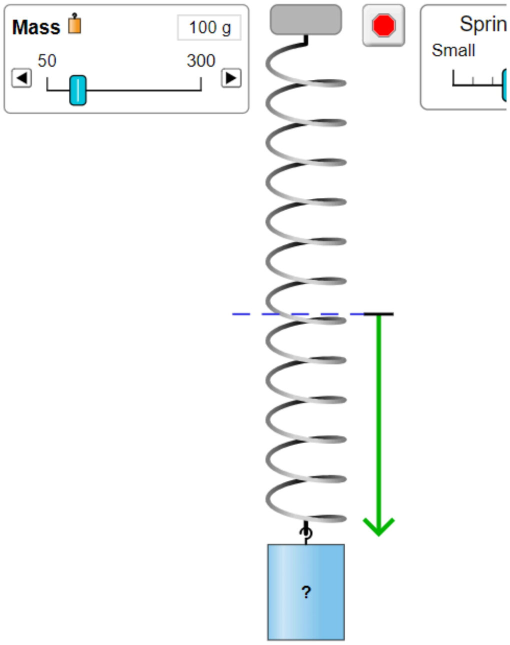
Hang the 100g (0.1kg) mass. click on the mass to stop it from moving. .

Use Hooke's law to find the spring constant. (weight is down and restoring force is up). Use kg and meters. Take g=9.81m/s/s. hint: measure the green arrow with the ruler. (slide 44)

- about 10N/m
- about 2 N/M
- about 4 N/m
- about 6 N/m

Question 7 14.5 pts

building on previous question. Now hang the blue unknown mass



Measure the displacement (stretch). That is the length of the green arrow. Using the K found in previous question. What is the mass of this blue mass?

- about 500g
- about 150g
- about 300g
- about 225g

Question 8 14.5 pts

bonus points.

The potential elastic energy stored in a spring stretched/ compressed by an amount x is:

PE = 0.5 K X^2

If you take the derivative you get:

- KX^2/6
- KX

- ○ K
- ○ KX^2

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